just turn them off for now, with *service iptables stop* and *setenforce 0*). Start the DNS service (*service named start*).

Configuring the slave DNS server

The first two steps are the same as for the master server, but the lines to be appended in *named.conf* will be:

```
zone "example.com " IN {
type slave;
file "slaves/example.com";
masters { 192.168.0.1; };
};
```

Again, adjust your firewall and SELinux settings or turn them off, and start the *named* service.

Now the master and slave DNS server is running. For the master machine, set the DNS server to 192.168.0.1, and for the slave server, set it to 192.168.0.2. Then verify the configuration by getting both machines to ping each other. If they are able to do so, then our master-slave DNS server configuration is working. On the slave server, check the */var/named/slaves* directory and you'll see a file *example.com*, which is the zone file for the *example.com* zone, transferred to the slave server from the master DNS server.

Security Pillar 1

It's time to switch to security, and the first pillar of this trilogy is transaction signatures (TSIG). First, run:

```
dnssec-keygen -a HMAC-MD5 -b 128 -n HOST server.example.com
```

You'll see two files generated, in the format *Kserver.example.com.+157+08837.key* and *Kserver.example.com.+157+08837.private*. The number 157 denotes the DNSSEC algorithm, and 08837 is the key's fingerprint, which will differ on your machine.

Open the *private* file and copy the key (you will find it in the third line of the file). On both machines, create a file */etc/server.key* with the following contents (replacing the secret string with the actual copied key):

```
key "server.example.com." {
     algorithm hmac-md5;
     secret "the key that you've copied";
};
```

Now run *chown root.named /etc/server.key* and on both servers, in *named.conf*, add *include "/etc/server.key"* at the top of the file. For the master server, add *allow-transfer { key server.example.com.; };* under the *options* field in *named.conf* while for the slave server's *named.conf*, add the following lines just below the new *include* line:

```
server 192.168.0.1 {
   keys { server.example.com.; };
   };
```

Delete the */var/named/slaves/example.com* file from the slave machine. Restart both the DNS servers and verify by pinging. You'll see that the *example.com* file appears again under */var/named/slaves*.

So far, as seen in our initial configuration, the *example.com* zone file is transferred to the slave DNS server via zone transfer. But under that configuration, zone files can be transferred to any machine. We have now implemented zone transfer security using TSIG in our first security drill—now, only machines with a valid key can get a zone file transferred from the master DNS server. You can verify this, if you like, by changing the slave's configuration. Again delete the *example.com* file under */var/named/slaves* and comment the include and server line in the slave's *named.conf* file. Restart the named service, and check the */var/named/slaves* directory. This time, you won't find the *example.com* file, because you're not using the key—so it is not being transferred.

Security Pillar 2

So far, we've seen the method of allowing only trusted machines to do a zone transfer. Now let's look at how to validate the authenticity of the zone file itself. On your master server, run the following commands:

```
dnssec-keygen -a DSA -b 512 -n ZONE example.com
dnssec-keygen -f KSK -a DSA -b 512 -n ZONE example.com
```

The first will create a Zone Signing Key (ZSK) in the familiar format *Kexample.com.003+46600.key* while the second will create a Key Signing Key (KSK) in the same format. Now open your zone file (forward) and append the following lines (the paths of the key files):

```
$include /var/named/ Kexample.com.+003+46600.key ;ZSK
$include /var/named/ Kexample.com.+003+44487.key ;KSK
```

Then run this command:

```
dnssec-signzone -o example.com \
   -k /var/named/Kexample.com.+003+44487.key \
   forward \
   /var/named/Kexample.com.+003+46600.key
```

This will create a file, *forward.signed*, which is the signed zone file. The -o switch specifies the zone name; *–k* the full path to the KSK file; and next is the name of the zone file (*forward*), followed by the path to the ZSK file.

Now change *named.conf* to use this signed file as the zone file; in the *file* line for the *zone "example.com"* block, change the name from *"forward"* to *"forward.signed"* so the block looks like what's shown below: